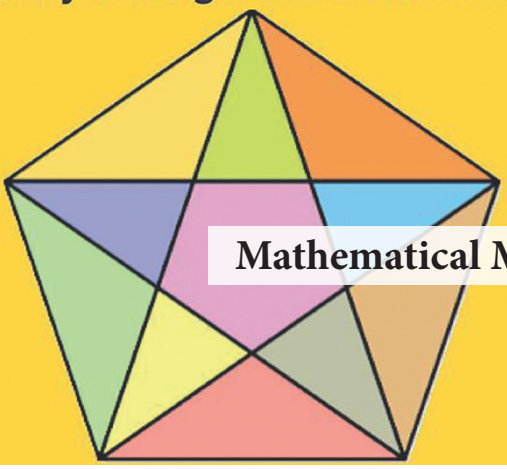
Chapter

7

How many triangles are there below?



Mathematical Methods



Srinivasa Ramanujan

(22 December, 1887 - 26 April, 1920)

Srinivasa Ramanujan, was an Indian Mathematician born on December 22, 1887, at Erode inTamilnadu, whose contributions to the **theory of numbers** include pioneering discoveries of the properties of the partition function. In 1911 Ramanujan published the first of his papers in the ***Journal of the Indian Mathematical Society.*** His genius slowly gained recognition, and in 1913 he began a correspondence with the British mathematician **Godfrey H. Hardy** that led to a special scholarship from the University of Madras and a grant from Trinity College, Cambridge.

Overcoming his religious objections, Ramanujan traveled to England in 1914, where Hardy tutored him and collaborated with him in some research.In England Ramanujan made further advances, especially in the **partition of numbers.** His papers were published in English and European journals, and in 1918 he was elected to **the Royal Society of London**.

In 1917 Ramanujan had contracted tuberculosis, but his condition improved sufficiently for him to return to India in 1919. He died the following year at Kumbakonam in Tamilnadu on April 26,1920, generally unknown to the world at large but recognized by mathematicians as a phenomenal genius. Ramanujan left behind three notebooks and a sheaf of pages (also called the **"lost notebook"**) containing many unpublished results that mathematicians continued to verify long after his death..

"Numbers are my friends".

- Srinivasa Ramanujan

BBNDWE

Learning Objectives

- Understands permutations and combinations
- Knows about the Binomial, Exponential and Logarithmic expansions
- Understands the concept of differentiation and applies in solving problems
- Understands the concept of integration and applies in solving problems